Exercise (2.1.1).

Proof. Notice that each $f \in A$ with nonzero constant term is a unit. Indeed, after multiplying by an element of k, we may assume that the constant term is 1. I.e. $f = 1 + gT$ for some $g \in A$. Then, since A is T-adically complete, we have that

$$\sum_{k=0}^{\infty}(-gT)^k$$

converges to some $h \in A$. But then $hf = h(1 + gT) = 1$ directly, so f is invertible.

It is also clear that $\{0A, TA\} \subseteq \operatorname{Spec}(A)$, and I claim this list is complete. Indeed, suppose $P \in \operatorname{Spec}(A)$ is nonzero. Then it contains some nonzero element which can be factorized as $T^r f$ for some f with nonzero constant term. But we've shown that f is a unit, so P contains T^r, and since it's prime, it contains T. But TA is maximal, so $P = TA$ as desired. □

Exercise (2.1.2).

Proof. Let $\mathfrak{m} \in \operatorname{Spec}(B)$ be a closed point, i.e. a maximal ideal of B. Then we can take the composition:

$$A \xrightarrow{\varphi} B \to B/\mathfrak{m}$$

Then $B/\mathfrak{m}$ is a finitely generated k-algebra and a field, so it's a finite field extension (Zariski Lemma). So, each element of A maps to an algebraic element over k. I.e. the image of A in $B/\mathfrak{m}$ is a finitely generated k-algebra that is algebraic over k. Thus, it must be a field, so $A/\varphi^{-1}(\mathfrak{m})$ is a field, whence $\varphi^{-1}(\mathfrak{m})$ is maximal. So, $(\operatorname{Spec}\varphi)(\mathfrak{m}) = \varphi^{-1}(\mathfrak{m})$ is a closed point. □

Exercise (2.1.3).

Proof. Since $A/\mathfrak{m}$ is a finite field extension of $\mathbb{R}$, it is either $\mathbb{R}$ or $\mathbb{C}$. So, it injects into $\mathbb{C}$, giving:

$$\mathbb{R}[X, Y] \to A \to A/\mathfrak{m} \hookrightarrow \mathbb{C}$$

Then the images of $1, x, x^2$ in $\mathbb{C}$ are linearly dependent over $\mathbb{R}$, so there is some $a, b, c \in \mathbb{R}$ with

$$a + bx + cx^2 \in \mathfrak{m}$$

If c is nonzero, dividing by it gives the desired quadratic in x contained in $\mathfrak{m}$. Otherwise, we must have b nonzero, so dividing by b and squaring gives the desired quadratic. The same argument applies to y. The same argument applied to $1, x, y$ gives the final claimed relation:

$$f = \alpha x + \beta y + \gamma \in \mathfrak{m}$$

and if $(\alpha, \beta) = (0, 0)$, then $\gamma \in \mathfrak{m}$ and $\gamma \neq 0$, which contradicts properness of $\mathfrak{m}$.

We'd like to show $\mathfrak{m} = fA$. WLOG, assume $\beta \neq 0$. It suffices to show that $(f, X^2 + Y^2 + 1)$ is maximal in $\mathbb{R}[X, Y]$, where we also use f to denote $\alpha X + \beta Y + \gamma$. We'll take the quotient iteratively. First, we claim that $\mathbb{R}[X, Y]/(f) \cong \mathbb{R}[t]$. Indeed, consider the map $\psi : \mathbb{R}[X, Y] \to \mathbb{R}[t]$ given by:

$$X \mapsto t \text{ and } Y \mapsto -\frac{\alpha t + \gamma}{\beta}$$

Clearly $f \in \ker\psi$ and ψ surjects. Further, if $g \in \ker\psi$, then by polynomial division (in $\mathbb{R}[X][Y]$), we can write:

$$g(X, Y) = f(X, Y)q(X, Y) + h(X)$$

since β is a unit. Thus $\psi(h) = 0$, but $\psi(h) = h(t)$ is zero in $\mathbb{R}[t]$ iff it is zero in $\mathbb{R}[X]$. I.e. $g = fq \in (f)$. So, $\mathbb{R}[X, Y]/(f) \cong \mathbb{R}[t]$, and under this isomorphism, X^2+Y^2+1 maps to a quadratic $p(t)$ (it is actually quadratic since the leading coefficient is positive, hence nonzero).

So, finally, to show that (f, X^2+Y^2+1) is maximal, it suffices to show that p is irreducible, i.e. that it has no root. Suppose it has a root u. Then we have the composite:

$$v : \mathbb{R}[X, Y] \to \mathbb{R}[t] \to \mathbb{R}$$

where the final map is evaluation $t \mapsto u$. Under the composition, we have $X^2 + Y^2 + 1 \mapsto p(u) = 0$, but this would mean that $0 = v(X)^2 + v(Y)^2 + 1 \geq 1$, which is a contradiction. Hence p is irreducible and $(f, X^2 + Y^2 + 1)$ is maximal, so that

$\mathfrak{m} = fA$.

The above proof shows in general that given $\alpha, \beta, \gamma \in \mathbb{R}$ with $(\alpha, \beta) \neq (0, 0)$, that $(\alpha x + \beta y + \gamma)$ is maximal in A. Multiplying the vector (α, β, γ) does not change this ideal, so we indeed get a map $\mathbb{P}^2_{\mathbb{R}} \setminus \{(0:0:1)\} \to \operatorname{Spec}(A)$. We've also shown that this map surjects on the set of maximal ideals. So, we only need to show it injects. In other words, suppose $\mathfrak{m} = (\alpha x + \beta y + \gamma) = (\alpha' x + \beta' y + \gamma')$ for some $(\alpha : \beta : \gamma) \neq (\alpha' : \beta' : \gamma'] \in \mathbb{P}^2_{\mathbb{R}} \setminus \{[0:0:1]\}$. We have two cases. First, suppose (α, β) and (α', β') are independent in $\mathbb{R}^2$. Then they form a basis, so we can find $a, b, c, d \in \mathbb{R}$ with:

$$\begin{pmatrix} \alpha & \alpha' \\ \beta & \beta' \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Then $\mathfrak{m}$ contains:

$$a(\alpha x + \beta y + \gamma) + c(\alpha' x + \beta' y + \gamma') = x + r$$

for a constant $r \in \mathbb{R}$. Similarly, $\mathfrak{m}$ contains:

$$b(\alpha x + \beta y + \gamma) + d(\alpha' x + \beta' y + \gamma') = y + s$$

for a constant $s \in \mathbb{R}$. But then, in $A/\mathfrak{m}$ we have $x \mapsto -r$ and $y \mapsto -s$, which gives:

$$0 = x^2 + y^2 + 1 \mapsto r^2 + s^2 + 1 \geq 1$$

which is a contradiction.

So we must have that (α, β) and (α', β') are linearly dependent. Since they are nonzero, there is a constant c with $(\alpha, \beta) = c(\alpha', \beta')$. We cannot have $\gamma = c\gamma'$, since the two points are distinct in $\mathbb{P}^2_{\mathbb{R}}$. But then $\mathfrak{m}$ contains:

$$(\alpha x + \beta y + \gamma) - c(\alpha' x + \beta' y + \gamma') = \gamma - c\gamma'$$

which is nonzero, contradicting the fact that $\mathfrak{m}$ is proper. So, we've exhausted all possibilities and must have that the map is injective. Thus, the association $(\alpha : \beta : \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ is a bijection between $\mathbb{P}^2_{\mathbb{R}} \setminus \{(0:0:1)\}$ and the maximal ideals of A.

As claimed, the map $\mathbb{R}[X] \to \mathbb{R}[X, Y] \to A$ is injective. Indeed, any element of the ideal $(X^2 + Y^2 + 1)$ necessarily has degree at least 2 in Y, while any polynomial in X has degree zero in Y. The morphism is also finite since we have a surjection of $\mathbb{R}[X]$-modules $\mathbb{R}[X]^2 \to A$ mapping (u, v) to $u + yv$. Indeed, given $g \in A$, we can lift it to $g(X, Y) \in \mathbb{R}[X, Y]$, and by polynomial division we can write

$$g(X, Y) = (X^2 + Y^2 + 1)q(X, Y) + Yr(X) + s(X)$$

so that after mapping into A we get:

$$g = yr(x) + s(x)$$

which is in the image of the $\mathbb{R}[X]$-module map above.

So now, if $\mathfrak{p}$ is a non-maximal prime of A, then I claim $\mathfrak{p} \cap \mathbb{R}[X]$ is a non-maximal prime of $\mathbb{R}[X]$. Indeed, let $\mathfrak{m}$ be a maximal ideal (strictly) containing $\mathfrak{p}$. Then, from the previous problem, $\mathbb{R}[X] \cap \mathfrak{m}$ is maximal and contains $\mathbb{R}[X] \cap \mathfrak{p}$. But we cannot have $\mathbb{R}[X] \cap \mathfrak{p} = \mathbb{R}[X] \cap \mathfrak{m}$ since A is an integral extension of $\mathbb{R}[X]$, which therefore satisfies incomparability. So $\mathbb{R}[X] \cap \mathfrak{p}$ is strictly contained in $\mathbb{R}[X] \cap \mathfrak{m}$, and therefore cannot be maximal.

But $\mathbb{R}[X]$ is a PID, so any nonzero prime is maximal, and therefore $\mathbb{R}[X] \cap \mathfrak{p} = 0$. Now let $g \in \mathfrak{p}$. Then $g \in A$, which is integral over $\mathbb{R}[X]$, so there is a relation:

$$g^n + a_{n-1}g^{n-1} + \cdots + a_0 = 0$$

where each $a_i \in \mathbb{R}[X]$. We may further assume that $n \geq 1$ is as small as possible. Now notice that a_0 is a multiple of g, so $a_0 \in \mathbb{R}[X] \cap \mathfrak{p} = 0$, so $a_0 = 0$. If $n > 1$, this gives

$$g^{n-1} + a_{n-1}g^{n-2} + \cdots + a_1 = 0$$

contradicting minimality of n. So instead we get $n = 1$, whence $g = -a_0 = 0$. So $\mathfrak{p} = 0$, and there is at most one non-maximal prime of A. $\square$

Exercise (2.1.4).

First, a geometric argument:

Proof. Notice that $\mathfrak{p}$ is an irreducible component of $\operatorname{Spec} A$. Then, if $f \in A_{\mathfrak{p}}$, then f is the germ of a function defined locally on $\mathfrak{p}$, and if $f \in \mathfrak{p}A_{\mathfrak{p}}$, then it vanishes on all of $\mathfrak{p}$. Hence, f vanishes everywhere, so by the Nullstellensatz, some power of it is zero.

Now, since A is reduced, only the zero function vanishes on all of $\operatorname{Spec} A$. Hence, if f is a zerodivisor, then there is some nonzero g with $fg = 0$, and g being nonzero implies that g does not vanish everywhere. Pick an irreducible component Z that g does not vanish on. Then

$$Z = Z \cap \operatorname{Spec}(A) = Z \cap V(fg) = Z \cap (V(f) \cup V(g)) = (Z \cap V(f)) \cup (Z \cap V(g))$$

By irreducibility, we must have $V(f) \supseteq Z$, so f vanishes on all of Z. □

Now, the formal algebra:

Proof. Note that $\mathfrak{p}A_{\mathfrak{p}}$ is the unique prime of $A_{\mathfrak{p}}$ since $\mathfrak{p}$ is minimal. Hence, the nilradical of $A_{\mathfrak{p}}$, as the intersection of the primes of $A_{\mathfrak{p}}$, is precisely $\mathfrak{p}A_{\mathfrak{p}}$. Hence, if $f \in \mathfrak{p}$ is nonzero, then its image is in $\mathfrak{p}A_{\mathfrak{p}}$, so there is some n such that $\frac{f^n}{1} = 0$. I.e. there is some $u \notin \mathfrak{p}$ with $f^n u = 0$, so f is a zerodivisor.

Now, let A be reduced, and let f be a zerodivisor, so there is some nonzero g with $fg = 0$. Then notice that g is not contained in every minimal prime, since otherwise it would be contained in

$$\bigcap_{\mathfrak{p} \text{ minimal prime}} \mathfrak{p} = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \sqrt{0} = 0$$

since A is reduced. So, there is some minimal prime $\mathfrak{p}$ with $g \notin \mathfrak{p}$. But $fg = 0 \in \mathfrak{p}$, so $f \in \mathfrak{p}$ by primality. □

Exercise (2.1.5).

Proof. Note that injectivity at the beginning and surjectivity at the end are obvious. So, we need to see exactness in the middle. Further, any multiple of $P_1(T_1)$ is clearly in the kernel, so the image is contained in the kernel. Finally, let $f \in k[T_1, \ldots, T_n]$ be in the kernel. Write it as a polynomial in $T_2, \ldots, T_n$ with coefficients in $k[T_1]$, as:

$$f = \sum_i a(T_1)T^i$$

where i runs over multi-indices and T^i denotes $T_2^{i_2} \cdots T_n^{i_n}$. Then, the image in $k_1[T_2, \ldots, T_n]$ is:

$$\sum_i \overline{a(T_1)}T^i$$

which must be zero. I.e. each coefficient must be zero, so $a(T_1) \in P_1(T_1)k[T_1]$. Thus, factoring out $P_1(T_1)$ from each summand shows that f is a multiple of $P_1(T_1)$.

Now, we induct on n. Indeed, $P_1(T_1) \subseteq \mathfrak{m}$, so $\mathfrak{m}$ corresponds to a maximal ideal $\mathfrak{m}_1$ of $k_1[T_2, \ldots, T_n]$. Let $p_2(T_2)$ generate $\mathfrak{m}_1 \cap k_1[T_2]$. Then, lift this back to a polynomial $P_2(T_1, T_2) \in k[T_1, T_2]$. Notice that if $f \in k[T_1, T_2] \cap \mathfrak{m}$, then the image is in $\mathfrak{m}_1 \cap k_1[T_2]$, so f is contained in (P_1, P_2). Let $k_2 = k_1[T_2]/(p_2)$ and continue in this way to get the desired sequence. □

Exercise (2.1.6).

Proof. Let Z be a closed subspace of a quasi-compact space X. Let $\{U_i\}$ be an open cover of Z. Then, each U_i is of the form $U_i' \cap Z$ for some U_i' open in X. Notice that $\{U_i'\} \cup \{X \setminus Z\}$ is an open cover of X. This has a finite subcover, say $U_1', \ldots, U_n', X \setminus Z$ (if this last set is not needed, it doesn't hurt to have it anyway). Then $U_1, \ldots, U_n$ is a cover of Z, so indeed any open cover of Z has a finite subcover. I.e. Z is quasi-compact.

Now, let $\{U_i\}$ be an open cover of $\operatorname{Spec} A$. By definition, each U_i is the complement of $V(J_i)$ for some ideal J_i of A. Then, we get:

$$V\left(\sum_i J_i\right) = \bigcap_i V(J_i) = \operatorname{Spec}(A) \setminus \left(\bigcup_i U_i\right) = \emptyset$$

I.e. this ideal is not contained in any prime ideal, and so cannot be proper. I.e.

$$\sum_i J_i = A$$

whence we have an equation:

$$\sum_i j_i = 1$$

where $j_i \in J_i$ and $j_i = 0$ for all but finitely many indices. But then if S is the support of this sum (the set of i where $j_i \neq 0$), then:

$$\sum_{i \in S} j_i = 1 \implies \sum_{i \in S} J_i = A$$

So:

$$\bigcup_{i \in S} U_i = \operatorname{Spec}(A) \setminus \bigcap_{i \in S} V(J_i) = \operatorname{Spec}(A) \setminus V\left(\sum_{i \in S} J_i\right) = \operatorname{Spec}(A) \setminus V(A) = \operatorname{Spec}(A)$$

so that we have found a finite subcover. I.e. $\operatorname{Spec} A$ is quasi-compact. □

Exercise (2.1.7).

Proof. It suffices to describe the irreducible closed subsets of $\operatorname{Spec} \mathbb{Z}[T]$, which are of the form $V(P)$ for $P \in \operatorname{Spec} \mathbb{Z}[T]$. Indeed, any closed set is then a finite union of these irreducible closed sets. We will organize this by $\operatorname{ht} P$:

The only height zero prime is (0), and $V(0) = \mathbb{A}^1_{\mathbb{Z}}$ is the whole space.

The height one primes are of the form (f) where $f \in \mathbb{Z}[T]$ is irreducible (over $\mathbb{Q}$, say), or of the form (p) for p a rational prime. I'm not sure what to say (geometrically) about $V(f)$, but note that $V(p) = \mathbb{A}^1_{\mathbb{F}_p}$.

Finally, the height two primes are of the form (f, p) for p a prime and f irreducible when reduced mod p (i.e. over $\mathbb{F}_p$). These are the closed points $V(f, p)$. □

Exercise (2.1.8).

Proof. In other words, we would like to show that if $\varphi : A \to B$ is integral and $\mathfrak{m} \in \operatorname{Spec} B$ is maximal, then $\varphi^{-1}\mathfrak{m}$ is maximal. Consider the induced map $\varphi A/\varphi^{-1}\mathfrak{m} \to B/\mathfrak{m}$. This is an injective integral extension (since the reduction of the same polynomials work) from a domain to a field. So, we must have that $A/\varphi^{-1}\mathfrak{m}$ is also a field, i.e. $\varphi^{-1}\mathfrak{m}$ is maximal as claimed.

The second claim is false. Indeed, if $\mathscr{O}$ is the ring of integers in a number field K and p is an unramified non-inert prime of the extension $\mathscr{O}/\mathbb{Z}$ lying over p, then the preimage $\operatorname{Spec}(\varphi)^{-1}(p\mathbb{Z})$ is the set of primes lying over p, which is not a singleton by choice. I.e. the preimage of a closed point is not a closed point (while $\mathbb{Z} \to \mathscr{O}$ is integral).

Directly, each element $b \otimes 1$ is integral over $A_{\mathfrak{p}}$ since it is integral over A and the same polynomial works. So, the integral closure of the image contains each simple tensor $b \otimes 1$ and each $1 \otimes (a/s)$, so it must be the full ring.

That T is multiplicative is obvious. To see that $T^{-1}B \neq 0$, it suffices to notice that $0 \notin T$ since φ is injective and $0 \notin A \setminus \mathfrak{p}$. So, lastly we show the isomorphism. Notice that we have an A-bilinear map $B \times A_{\mathfrak{p}} \to T^{-1}B$ given by:

$$\left(b, \frac{a}{s}\right) \mapsto \frac{\varphi(a)b}{\varphi(s)}$$

which thus factors through a map $f : B \otimes_A A_{\mathfrak{p}} \to T^{-1}B$. We also have a map $B \to B \otimes_A A_{\mathfrak{p}}$ given by $b \mapsto b \otimes 1$. If $t = \varphi(a) \in T$, where $a \notin \mathfrak{p}$, then it maps to $t \otimes 1$, which satisfies:

$$(t \otimes 1)\left(1 \otimes \frac{1}{a}\right) = \left(\varphi(a) \otimes \frac{1}{a}\right) = a \cdot \left(1 \otimes \frac{1}{a}\right) = 1 \cdot 1$$

so that each element of T maps to something invertible. So, the map factors through a map $g : T^{-1}B \to B \otimes_A A_{\mathfrak{p}}$. These are the desired isomorphisms. Indeed, for $b/t \in T^{-1}B$:

$$f(g(b/t)) = f((b \otimes 1)(t \otimes 1)^{-1}) = \frac{b}{1}\left(\frac{t}{1}\right)^{-1} = b/t$$

so this composition is the identity. For a simple tensor $b \otimes (a/s)$, we have:

$$g(f(b \otimes (a/s))) = g\left(\frac{\varphi(a)b}{\varphi(s)}\right) = (\varphi(a)b \otimes 1)(\varphi(s) \otimes 1)^{-1} = a(b \otimes 1)[s(1 \otimes 1)]^{-1} = a(b \otimes 1)(1 \otimes (1/s)) = b \otimes (a/s)$$

and so is identity on simple tensors, and hence on all of the domain. So, f and g do indeed establish the isomorphism.

This gives the final claim. Indeed, given $\mathfrak{p} \in \operatorname{Spec} A$, we have the square:

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ A_{\mathfrak{p}} \arrow[r] & B \otimes_A A_{\mathfrak{p}} = T^{-1}B \end{tikzcd}$$

Let $\mathfrak{m}$ be any closed point in $\operatorname{Spec} T^{-1}B$. Then, the image along the bottom arrow is a closed point in $\operatorname{Spec} A_{\mathfrak{p}}$, which therefore must be $\mathfrak{p}A_{\mathfrak{p}}$, as this is the unique closed point. Taking the further image along the localization map thus gives the ideal $\mathfrak{p} \in \operatorname{Spec} A$. But since this diagram commutes, the image in $\operatorname{Spec} B$ of $\mathfrak{m}$ gives a point of $\mathfrak{m}' \in \operatorname{Spec} B$ such that $\operatorname{Spec}(\varphi)(\mathfrak{m}') = \mathfrak{p}$ as desired. $\square$

Exercise (2.1.9).

Proof. First, note that since A is a finite k-module, the extension is integral. Let $\mathfrak{p} \in \operatorname{Spec} A$; then it is contained in some maximal ideal $\mathfrak{m} \in \operatorname{Spec} A$. Then $\mathfrak{p} \cap k$ and $\mathfrak{m} \cap k$ must be primes in k, i.e. they must both be zero. So, they lie over the same prime, and since integral extensions satisfy incomparability, the containment $\mathfrak{p} \subseteq \mathfrak{m}$ implies $\mathfrak{p} = \mathfrak{m}$. So, indeed every prime is maximal.

Now, suppose $\operatorname{Spec} A$ is has at least $d+1$ elements for $d = \dim_k(A)$. Then choose distinct prime (maximal) ideals $\mathfrak{m}_1, \mathfrak{m}_2, \mathfrak{m}_3, \ldots, \mathfrak{m}_{d+1}$. Let

$$I_r = \bigcap_{j=1}^{r} \mathfrak{m}_j$$

Then we have the descending chain:

$$A \supseteq I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots I_{d+1}$$

Each of these containments must be proper. Indeed, otherwise we would have

$$\mathfrak{m}_1 \cdots \mathfrak{m}_r \subseteq I_r \subseteq \mathfrak{m}_{r+1}$$

But then, since $\mathfrak{m}_{r+1}$ is prime, it contains some $\mathfrak{m}_i$, whence $\mathfrak{m}_{r+1} = \mathfrak{m}_i$ by maximality, contrary to assumption.

On the other hand, each of these is a k-submodule of A, i.e. a k-vector space. Since the containments are proper, we must have that $\dim_k(I_{r+1}) < \dim_k(I_r)$. But $\dim_k(A) = d$, so this is a contradiction, as we cannot have a strictly decreasing sequence of length $d+2$ among the integers $\{0, \ldots, d\}$. Thus, we must have $|\operatorname{Spec} A| \leq \dim_k(A)$.

Note that $k[T_1, \ldots, T_d]$ is a UFD, and a nonzero polynomial is a unit iff it is a constant. Hence, Euclid's proof shows that there are infinitely many prime elements and so infinitely many primes. Explicitly: note that T_1 is prime, so there is at least one nonzero prime. Suppose that $\pi_1, \ldots, \pi_n$ is a list of nonzero primes, so that none of them may be constant. Then $a = \pi_1 \cdots pi_n + 1$ factors as a nonempty product of primes since it has positive degree. None of these primes can be any π_i, else π_i divides 1. So, no finite list of primes of positive degree is complete, hence there are infinitely many primes.

We've already shown one direction, so now suppose that A is not a finite k-module. By Noether normalization, we can find a finite injective morphism $k[T_1, \ldots, T_d] \to A$. We must have $d \geq 1$ since we've assumed A is not finite over k. But now, the extension A over $k[T_1, \ldots, T_d]$ satisfies lying over, so we have at least one prime of A over each prime of $k[T_1, \ldots, T_d]$. In fact, this association is injective since each prime of A lies over a unique prime of $k[T_1, \ldots, T_d]$. So, $|\operatorname{Spec} A| \geq |\operatorname{Spec} k[T_1, \ldots, T_d]| = \infty$ as desired. $\square$

Exercise (2.2.1).

Proof. Following the explicit construction given, the sheafification of the constant presheaf is the sheaf $\mathscr{A}^\dagger$ where $\mathscr{A}^\dagger(U)$ is the set of functions $f : U \to \bigsqcup_{x \in U} \mathscr{A}_x$ given locally by sections. Clearly $\mathscr{A}_x = A$, and the fact that f is locally given by sections means that for each $x \in U$, there is some open V and $a \in \mathscr{A}(V) = A$ with $f(x) = a_x = a$ for each $x \in V$. Another way to state this is that $f : U \to A$ is continuous when A is given the discrete topology. I.e. $\mathscr{A}^\dagger$ is the sheaf of continuous functions from subsets of X to A with the discrete topology, i.e. locally constant functions $X \to A$. Finally, we can interpret this as $\mathscr{A}^\dagger(U)$ being the direct product of copies of A in bijection with the connected components of U.

If every open subset of X is connected, then this description shows that $\mathscr{A} = \mathscr{A}^\dagger$, so $\mathscr{A}$ is a sheaf. Conversely, if there is some $U \subseteq X$ that is the disjoint union of open subsets V, W of X, then $\mathscr{A}$ is not a sheaf. Indeed, A is nontrivial, so we can choose two distinct elements $a, b \in A$. Then $a \in \mathscr{A}(V)$ and $b \in \mathscr{A}(W)$, and these sections agree on overlaps since $V \cap W = \emptyset$, but there is no $x \in \mathscr{A}(U)$ with $x|_V = x = a$ and $x|_W = x = b$. So, $\mathscr{A}$ does not satisfy gluing. $\square$

Exercise (2.2.2).

Proof. Let $A = \{x \in X : s_x = t_x\}$. Suppose that $x \in A$. Then there is an open neighborhood U of x with $s|_U = t|_U$. But then for any $y \in U$, we have

$$s_y = (s|_U)_y = (t|_U)_y = t_y$$

so $y \in A$. Thus $U \subseteq A$ and we've shown that each $x \in A$ is contained in an open set contained in A, so A is open as claimed. □

Exercise (2.2.3).

Proof. I'll come back to this one. □

Exercise (2.2.4).

Proof. Name the maps:

$$0 \to \mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}'' \to 0$$

Recall that this implies that for each $x \in X$ that we have an exact sequence:

$$0 \to \mathscr{F}'_x \xrightarrow{\varphi_x} \mathscr{F}_x \xrightarrow{\psi_x} \mathscr{F}''_x \to 0$$

Now, exactness at $\mathscr{F}'(X)$ is obvious, since a map of sheaves is injective iff it is injective on each open set. So, we would like to show exactness at $\mathscr{F}(X)$. First, let $s \in \mathscr{F}'(X)$. Then for each $x \in X$ we have:

$$[\psi(X)(\varphi(X)(s))]_x = \psi_x([\varphi(X)(s)]_x) = \psi_x(\varphi_x(s_x)) = 0$$

since $\psi_x \circ \varphi_x = 0$. Hence the image of $\varphi(X)$ is contained in the kernel of $\psi(X)$. Conversely, suppose $s \in \ker(\psi(X)) \subseteq \mathscr{F}(X)$. Then, again, for each x, we have

$$\psi_x(s_x) = [\psi(X)(s)]_x = 0$$

So, $s_x \in \ker(\psi_x) = \operatorname{im}(\varphi_x)$. Hence, for each x we can find an open neighborhood $U(x)$ of x and a section $t(x) \in \mathscr{F}'(U(x))$ with $\varphi(U(x))(t(x)) = s|_{U(x)}$. We would like to glue these to a single global section, and so we need to see that they agree on overlaps.

To that end, let $x, y \in X$, let $a = t(x)|_{U(x)\cap U(y)}$, and let $b = t(y)|_{U(x)\cap U(y)}$. If $U(x) \cap U(y) = \emptyset$, then we must have $a = b$, and there is nothing to show. Otherwise, let $z \in U(x) \cap U(y)$. Then we have:

$$\varphi_z(a_z) = \varphi_z(t(x)_z) = [\varphi(U(x))(t(x))]_z = (s|_{U(x)})_z = s_z$$

and similarly:

$$\varphi_z(b_z) = \varphi_z(t(y)_z) = [\varphi(U(y))(t(y))]_z = (s|_{U(y)})_z = s_z$$

So, $\varphi_z(a_z) = \varphi_z(b_z)$ and by injectivity of φ_z we get $a_z = b_z$. Since this is true for all $z \in U(x) \cap U(y)$, we get that $a = b$, and so the $t(x)$ do indeed agree on overlaps. Hence they glue to a single section $t \in \mathscr{F}'(X)$ with $t|_{U(x)} = t(x)$.

But then we are finished, for if $x \in X$, then:

$$[\varphi(X)(t)]_x = \varphi_x(t_x) = \varphi_x((t|_{U(x)})_x) = \varphi_x(t(x)_x) = [\varphi(U(x))(t(x))]_x = (s|_{U(x)})_x = s_x$$

and so $\varphi(X)(t) = s$ since they are equal on all stalks. I.e. s is in the image of $\varphi(X)$. □

Exercise (2.2.5).

Proof. □

Exercise (2.2.6).

Proof. □

Exercise (2.2.7).

Proof. □

Exercise (2.2.8).

Proof. □

Exercise (2.2.9).

Proof. □

Exercise (2.2.10).

Proof. □

Exercise (2.2.11).

Proof. □

Exercise (2.2.12).

Proof. □

Exercise (2.2.13).

Proof. □

Exercise (2.2.14).

Proof. □

Exercise (2.3.1).

Proof. □

Exercise (2.3.2).

Proof. Let $i : U \to X$ be the inclusion map, so that we wish to show $i^\# : A \to B$ is flat. It suffices to show this locally. That is, for $\mathfrak{q} \in \operatorname{Spec} B = U$ and $\mathfrak{p} = i(\mathfrak{q}) \in \operatorname{Spec} A$, we wish to show that $A_\mathfrak{p} \to B_\mathfrak{q}$ is flat. But this is the map $i^\#_\mathfrak{q} : U_\mathfrak{q} \to X_{i(\mathfrak{q})}$, which is an isomorphism since U is an open subscheme. Isomorphisms are clearly flat, so the claim is shown. □

Exercise (2.3.3).

Proof. Notice that if $\mathfrak{p} \in F$, then $\mathfrak{p} \supseteq I$ tautologically, and so $\mathfrak{p} \in V(I)$ by definition. I.e. we immediately have $F \subseteq V(I)$, and since $V(I)$ is closed, this gives $\overline{F} \subseteq V(I)$. We wish to show the reverse.

Conversely, we have that $\overline{F}$ is a closed set, so it is of the form $V(J)$ for some ideal J. Then:

$$J \subseteq \sqrt{J} = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{\mathfrak{p} \in \overline{F}} \mathfrak{p} \subseteq \bigcap_{\mathfrak{p} \in F} \mathfrak{p} = I$$

where the inclusion comes from the fact that we are intersecting over a subset. So, by applying V, we get: $\overline{F} = V(J) \supseteq V(I)$, as desired.

Any prime $\mathfrak{p} \in \operatorname{im}(f)$ contains $\ker \varphi$, so $\operatorname{im}(f) \subseteq V(\ker \varphi)$, and since the latter is closed, we have $\overline{\operatorname{im}(f)} \subseteq V(\ker \varphi)$. Conversely, again write $\overline{\operatorname{im}(f)}$ as $V(J)$ for some $J \in \operatorname{Spec} A$. Then, for any prime $\mathfrak{p} \in \operatorname{Spec} B$, we have $f(\mathfrak{p}) \in V(J)$, i.e. $\varphi^{-1}\mathfrak{p} \supseteq J$. Thus, $\varphi(J) \subseteq \mathfrak{p}$. Since this is true of any prime, $\varphi(J)$ is contained in their intersection, i.e. $\varphi(J) \subseteq \sqrt{0}$, and so $J \subseteq \sqrt{\ker \varphi}$. But then $V(J) \supseteq V(\ker \varphi)$ as desired.

When we have $\varphi : A \to A_\mathfrak{p}$ for some $\mathfrak{p} \in \operatorname{Spec} A$, then on the one hand $\operatorname{im} f$ is precisely the set of primes contained in $\mathfrak{p}$. On the other hand, we have $\ker \varphi = \{a \in A : au = 0 \text{ for some } u \notin \mathfrak{p}\} \subseteq \mathfrak{p}$. So, there is some subideal I of $\mathfrak{p}$ such that the lattice of ideals containing I corresponds (bijectively, inclusion-preserving) to the closure of the lattice of ideals contained in $\mathfrak{p}$. □

Exercise (2.3.4).

Proof. □

Exercise (2.3.5).

Proof. Let $A = \mathscr{O}_Y(Y)$ be the global sections on Y and let $X = \operatorname{Spec} A$. We will show that $Y \cong X$, and hence is affine as claimed.

Notice that $\mathscr{O}_X(X) = A$. In particular, we have the identity ring homomorphism $i : \mathscr{O}_Y(Y) \to \mathscr{O}_X(X)$, and this gives rise to a morphism $f : X \to Y$ of schemes with $\rho(f) = i$ since ρ is surjective by assumption. Similarly, since X is affine, we can apply Proposition 3.25 directly (with the roles of X and Y reversed). That is, for the identity map $j : \mathscr{O}_X(X) \to \mathscr{O}_Y(Y)$, we get a morphism $g : Y \to X$ with $\rho(g) = j$ since ρ is also surjective in this case. Now, we use functoriality of ρ. Namely, we have:

$$\rho(g \circ f) = \rho(f) \circ \rho(g) = i \circ j = \mathrm{id}_{\mathscr{O}_X(X)} = \rho(\mathrm{id}_X)$$

But since X is affine, ρ is also injective, so we get that $g \circ f = \mathrm{id}_X$. Similarly, we have

$$\rho(f \circ g) = \rho(g) \circ \rho(f) = j \circ i = \mathrm{id}_{\mathscr{O}_Y(Y)} = \rho(\mathrm{id}_Y)$$

However, we do not automatically know that ρ is injective in this case: the assumption only gives us this for maps $X \to Y$ with X affine, while we have a map $Y \to Y$ and are trying to show Y is affine. So, instead, let us work locally. Namely, let $U \subseteq Y$ be an affine chart, and let $\iota : U \hookrightarrow Y$ be the inclusion. Then,

$$\rho(f \circ g \circ \iota) = \rho(\iota) \circ \rho(g) \circ \rho(f) = \rho(\iota) \circ \mathrm{id}_{\mathscr{O}_Y(Y)} = \rho(\iota)$$

But now, we do have injectivity, since U is affine. I.e. we get $\iota = f \circ g \circ \iota$. Written alternatively, we have that $f \circ g$ and the identity map on Y agree when restricted to any affine chart U. Since these form a cover, we get that these global sections are equal everywhere, i.e. $f \circ g = \mathrm{id}_Y$. Thus, we have shown that f, g furnish an isomorphism between X and Y as desired. □

Exercise (2.3.6).

Proof. □

Exercise (2.3.7).

Proof. Note that $x \in X(k)$ means that $x : \operatorname{Spec} k \to X$ is a section of the structure morphism $X \to \operatorname{Spec} k$. Thus, $f(x)$ is shorthand for $f \circ x : \operatorname{Spec} k \to \mathbb{A}^n_k$. The identification of $\mathbb{A}^n_k(k)$ with k^n is as follows: given a section $t : \operatorname{Spec} k \to \mathbb{A}^n_k$, we get a map $t^\# : k[T_1, \ldots, T_n] \to k$, and we identify t with the tuple $(t^\#(T_1), \ldots, t^\#(T_n))$. So, in our case, $f(x)$ corresponds to the point whose ith coordinate is:

$$(f \circ x)^\#(T_i) = x^\#(f^\#(T_i)) = x^\#(\varphi(T_i)) = x^\#(f_i) = f_i(x)$$

as claimed. □

Exercise (2.3.8).

Proof. The first claim is immediate. Indeed, if $x \in X$, then since the $f_{i,x}$ generate $\mathscr{O}_{X,x}$, at least one of them cannot be in the maximal ideal of this local ring. That is, $f_i \notin \mathfrak{m}_{X,x}$ for some i. But then $x \in X_{f_i}$ for this i, so X is the union of the X_{f_i}.

Now, note that if $f \in \mathscr{O}_X(X)$ for any scheme X, then there is some $g \in \mathscr{O}_X(X_f)$ with $(f|_{X_f})g = 1$. That is, $f|_{X_f}$ is a unit. This is clear: for $p \in X_f$, we have f_p is invertible, so there is some open U containing p and some $g(p) \in \mathscr{O}_X(U)$ with $(f|_U)g(p) = 1$. Gluing these gives the g we seek.

For $i \in \{0, \ldots, n\}$, let $A_i = A[u_{ij}]$ for variables u_{ij} with $j \in \{0, \ldots, n\} \setminus \{i\}$. That is, each A_i is a polynomial ring over A in n variables. Define the ring homomorphisms $\varphi_i : A_i \to \mathscr{O}_{X_f}(X_f)$ by $\varphi_i(u_{ij}) = (f_i|_{X_{f_i}})^{-1}(f_j|_{X_{f_i}})$. By the note above, this is well-defined, since $f_i|_{X_{f_i}}$ is invertible. These induce morphisms $F_i : X_{f_i} \to \operatorname{Spec} A_i$.

Notice that

$$D_+(T_i) = \operatorname{Spec} A[T_0, \ldots, T_n]_{(T_i)} = \operatorname{Spec} A\left[\frac{T_0}{T_i}, \ldots, \frac{T_n}{T_i}\right]$$

which has a clear isomorphism with $\operatorname{Spec} A_i$ given by $u_{ij} \mapsto T_j/T_i$. Via these isomorphisms, we can extend our maps to $F_i : X_{f_i} \to \mathbb{P}^n_A$ by composing with the inclusion maps.

Finally, we will glue these maps to get a single morphism $f : X \to \mathbb{P}^n_A$. It is clear from the construction that f satisfies the desired properties.

In the case that A is a field and $x \in X(A)$ is a rational point, we can write $f_i(x)$ for the image of $f_{i,x}$ in $\mathscr{O}_{X,x}/\mathfrak{m}_{X,x} = A$. Then, via the identification of the A-rational points of $\mathbb{P}^n_A$ with equivalence classes of $(n+1)$-tuples in A, we have $f(x) = [f_0(x) : \cdots : f_n(x)]$. □

Exercise (2.3.9).

Proof. □

Exercise (2.3.10).

Proof. Let $f \in \mathscr{O}_X(X)$ be a global section, and let $U_i = D_+(T_i)$ so that $U_i \cong \operatorname{Spec} A[T_0/T_i, \ldots, T_n/T_i]$. Fix some indices $i \neq j$. Then we have that the restriction of f to U_i can be expressed as a polynomial in the variables $T_0/T_i, \ldots, T_n/T_j$, so after clearing denominators, we have:

$$g = T_i^a (f|_{U_i})$$

for some $g \in A[T_0, \ldots, T_n]$ that is not a multiple of T_i. But similarly, we can clear denominators in j, i.e.

$$T_j^b (f|_{U_j}) \in A[T_0, \ldots, T_n]$$

for some b. In the intersection, this implies:

$$T_j^b g = T_i^a \left(T_j^b (f|_{U_i \cap U_j})\right)$$

which shows that $T_j^b g$ is a multiple of T_i^a in $A[T_0, \ldots, T_n]$. By uniqueness of polynomial representation, we must have $a = 0$ since T_i divides neither T_j nor g. But this means that $f|_{U_i} \in A$. Since i is arbitrary, this shows that $f \in A$, and since f was arbitrary, this gives that $\mathscr{O}_X(X) = A$.

Now, note that if $n = 0$, then $\mathbb{P}^n_A = \operatorname{Spec} A$ is affine. Conversely, suppose that $\mathbb{P}^n_A$ is affine, so it is isomorphic to $\operatorname{Spec} B$ for some ring B. But then $A = \mathscr{O}_X(X) \cong \mathscr{O}_{\operatorname{Spec} B}(\operatorname{Spec} B) = B$, so we have $\mathbb{P}^n_A \cong \operatorname{Spec} A$. We want to show that this implies $n = 0$. One approach (using material from later sections) is to reduce to the case for fields. Namely, choosing an arbitrary maximal ideal $\mathfrak{m}$ of A (i.e. a closed point) with residue field $k = A/\mathfrak{m}$ gives the following fibre product:

$$\begin{tikzcd}
\mathbb{P}^n_k \arrow[r] \arrow[d] & \mathbb{P}^n_A \arrow[d] \\
\operatorname{Spec} k \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

where the right edge is the claimed isomorphism. But then the left arrow would also be an isomorphism, which is easily disproved by counting. That is, an isomorphism of schemes induces in particular a homeomorphism of topological spaces, but $\operatorname{Spec} k$ consists of a single point, while $\mathbb{P}^n_k$ has at least two points for $n > 0$: both the zero ideal and the ideal generated by T_0 are homogeneous, prime, and do not contain the irrelevant ideal. So, we cannot have $n > 0$ and we conclude $n = 0$ as desired. □

Exercise (2.3.11).

Proof. □

Exercise (2.3.12).

Proof. □

Exercise (2.3.13).

Proof. □

Exercise (2.3.14).

Proof. □

Exercise (2.3.15).

Proof. □

Exercise (2.3.16).

Proof. □

Exercise (2.3.17).

Proof. □

Exercise (2.3.18).

Proof. □

Exercise (2.3.19).

Proof. □

Exercise (2.3.20).

Proof. □

Exercise (2.3.21).

Proof. □

Exercise (2.3.22).

Proof. □

Exercise (2.4.1).

Proof. □

Exercise (2.4.2).

Proof. □

Exercise (2.4.3).

Proof. □

Exercise (2.4.4).

Proof. □

Exercise (2.4.5).

Proof. □

Exercise (2.4.6).

Proof. □

Exercise (2.4.7).

Proof. □

Exercise (2.4.8).

Proof. □

Exercise (2.4.9).

Proof. □

Exercise (2.4.10).

Proof. □

Exercise (2.4.11).

Proof. □

Exercise (2.4.12).

Proof. □

Exercise (2.5.1).

Proof. □

Exercise (2.5.2).

Proof. □

Exercise (2.5.3).

Proof. □

Exercise (2.5.4).

Proof. □

Exercise (2.5.5).

Proof. □

Exercise (2.5.6).

Proof. □

Exercise (2.5.7).

Proof. □

Exercise (2.5.8).

Proof. □

Exercise (2.5.9).

Proof. □

Exercise (2.5.10).

Proof. □

Exercise (2.5.11).

Proof. □

Exercise (2.5.12).

Proof. □

Exercise (2.5.13).

Proof. □

Exercise (2.5.14).

Proof. □

Exercise (2.5.15).

Proof. □